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ORIGINAL PAPER

Race/ethnicity and changing US socioeconomic gradients in breast cancer incidence: California and Massachusetts, 1978–2002 (United States)

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Abstract
Objective We tested the hypothesis that the US socioeconomic gradient in breast cancer incidence is declining, with the decline most pronounced among racial/ethnic groups with the highest incidence rates.
Methods We geocoded the invasive incident breast cancer cases for three US population-based cancer registries covering: Los Angeles County, CA (1978–1982, 1988–1992, 1998–2002; $n = 68{,}762$ cases), the San Francisco Bay Area, CA (1978–1982, 1988–1992, 1998–2002; $n = 37{,}210$ cases) and Massachusetts (1988–1992, 1998–2002; $n = 48{,}111$ cases), linked the records to census tract area-based socioeconomic measures, and, for each socioeconomic stratum, computed average annual breast cancer incidence rates for the 5-year period straddling the 1980, 1990, and 2000 census, overall and by race/ethnicity and gender.
Results Our findings indicate that the socioeconomic gradient in breast cancer incidence is: (a) relatively small (at most 1.2) and stable among US white non-Hispanic and black women; (b) sharper and generally increasing among Hispanic and Asian and Pacific Islander American women; and (c) cannot be meaningfully analyzed without considering effect modification by race/ethnicity and immigration.
Conclusion Our results indicate that secular changes in US socioeconomic gradients in breast cancer incidence exist and vary by race/ethnicity.

Keywords Breast cancer incidence · Epidemiology · race/ethnicity · Socioeconomic position · Surveillance

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Introduction

Epidemiologic evidence indicates that for much of the 20th century CE, the incidence of breast cancer has, on average, been higher among women with more versus less socioeconomic resources [1–3]. The socioeconomic patterning of a disease, however, is not necessarily fixed, but instead depends on the relative mix of both changing and persistent pathways linking economic resources to health status [3]. For example, in the case of lung cancer, 20th century shifts in the socioeconomic distribution of smoking – from being less to more associated with economic deprivation – led to corresponding changes in the socioeconomic distribution of lung cancer incidence and mortality [4].

Raising the possibility of similar shifts in breast cancer epidemiology, during the 20th century the social patterning of important breast cancer risk factors – include age at menarche, parity and age at first childbirth, and also height and weight [1–3] – has likewise changed. Data from the UK indicate, for example, that between 1910 and 1960, the average age at menarche dropped 4.4 months per decade among working class girls, due to improved nutrition and a decline in childhood infectious disease, compared to a drop of 2.8 months per decade among middle class girls [5]. By mid-century, the rate of decline in the age at menarche and increase in adult height began to level, and concomitantly, the socioeconomic gradient for each also diminished, largely due to relatively poorer sectors of the population ''catching up.''[5, 6] Additionally, during the latter part of the 20th century, in the US a shift to lower parity and a higher rate of first births after age 30 became evident among women across all educational levels [7]. Concomitantly, the late 20th century rise in childhood obesity, especially among working class and poor populations, [8] in turn has led – given links between childhood obesity and earlier onset of menarche [9] – to documented reversals in the socioeconomic gradient in age at menarche, with earlier onset now observed among poorer and heavier girls [5, 10]. Adult obesity is likewise rising most quickly among economically deprived populations, with research indicating that postmenopausal obesity may increase risk of breast cancer [2, 11].

Together, these secular changes in the socioeconomic distribution of breast cancer risk factors would lead to the prediction of a faster rise in breast cancer incidence among less versus more affluent women and to a corresponding decline of the socioeconomic gradient in breast cancer incidence. To date, however, only a relatively small numbers of population-based studies, in the US [12–22] and elsewhere [23–30] have attempted to quantify the socioeconomic gradient in breast cancer incidence, using a variety of socioeconomic measures, either at a given point in time or, in a handful of cases, over time [17, 28]. Considered together, these limited data suggest that over the course of the 20th century, in both the US and other affluent industrialized countries, the breast cancer socioeconomic gradient appears to have diminished, dropping from a relative risk exceeding 2.0 down to one nearly equal to 1.0, comparing the most to least affluent women, with this shift largely due to rates catching up among less affluent women [31] If true, such a decline would have major implications for not only public perceptions of and policies to address breast cancer, but also the population burden of breast cancer mortality, given strong associations between economic deprivation and shortened survival once diagnosed [1, 22].

We accordingly sought to test the hypothesis that the magnitude of US socioeconomic gradients in breast cancer is decreasing over time. To do this, we obtained US population-based cancer registry data for the period 1978 through 2002, which we geocoded and linked to census tract socioeconomic data [21, 32]. Our *a priori* hypothesis, based on prior research, [31] was that the magnitude of the decrease in the socioeconomic gradient would vary by race/ethnicity, with the decline most evident among groups with the highest overall rates of breast cancer, i.e., US white non-Hispanic and black women [22, 31].

Materials and methods

Study population

We obtained all cases of primary invasive breast cancer, excluding *in situ* carcinoma, occurring between 1978–1982, 1998–1992, and 1998–2002 and recorded by August 2004 in three population-based cancer registries: (1) the Northern California Cancer Center's (NCCC) San Francisco/Oakland SEER cancer registry, encompassing five counties (Alameda, Contra Costa, Marin, San Francisco, and San Mateo; n = 37,210), [33] (2) the Los Angeles Cancer Surveillance Program (LA CSP), encompassing Los Angeles County (n = 68,762), [34] and (3) the Massachusetts Cancer Registry (MCR), encompassing the entire state (for 1988–1992 and 1998–2002 only, since the registry was not established until 1982; n = 48,111) [35]. We chose these registries because their catchment areas were sufficiently large and heterogeneous for socioeconomic position and race/ethnicity (Figures 1 and 2) to enable us to test our hypotheses. Additionally, all three registries have an excellent track record regarding completeness (estimated at >98%), timeliness, and accuracy of registering cancer cases [33–35]. We retained individuals with multiple primary tumors and excluded the small number of cases categorized as transgender or of unknown gender (since we had no corresponding available denominator data; n = 19), and with unknown age (since they could not be included in age-standardized analyses; n = 79). Use of these data for our investigation was approved by each registry's Institutional Review Board and by the Harvard School of Public Health Human Subjects Committee.

We obtained population counts at the census tract level by age, gender, and census-defined race (white; black; Asian and Pacific Islander; American Indian and Alaskan Native) and Hispanic ethnicity from the 1980 (Summary Tape File 2 [STF2]), 1990 (STF2), and 2000 (Summary File 2 [SF2]) US Census data [36]. Population counts by age and gender were available for the white non-Hispanic

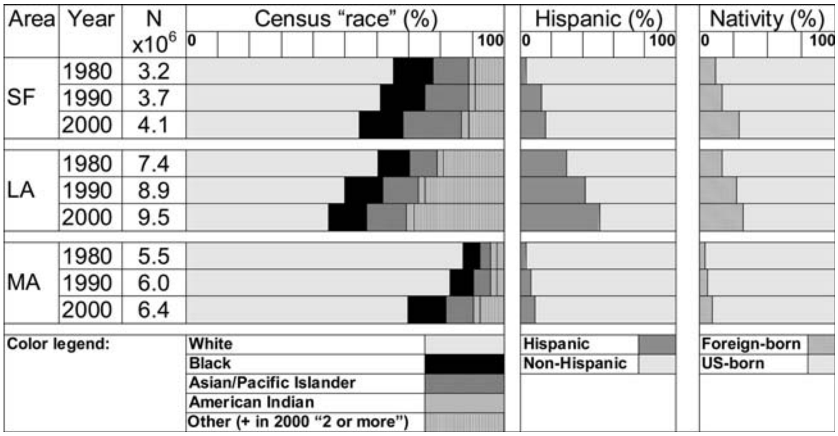

Fig. 1 Study catchment areas: sociodemographic composition by census-defined ''race,'' Hispanic ethnicity, and nativity for San Francisco Bay Area (SF) and Los Angeles County (LA), California, and Massachusetts (MA), 1980, 1990, and 2000 census [45]

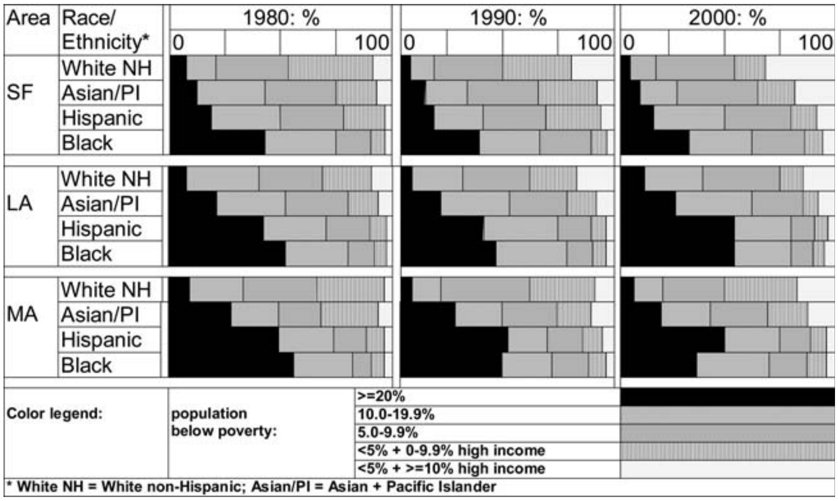

Fig. 2 Study catchment areas: racial/ethnic distribution by census tract poverty, for San Francisco Bay Area (SF) and Los Angeles County (LA), California, and Massachusetts (MA), 1980, 1990, and 2000 census [45]

population only for the 1990 and 2000 census; for 1980, we imputed these data by applying the proportion of non-Hispanic whites (derived from the 1980 STF3) to the white age and gender population estimates in 1980 STF2. To estimate person-time denominators, we multiplied the decennial population counts by 5, thereby enabling estimation of average annual incidence rates, which we age-standardized to the year 2000 standard million [37].

We conducted analyses by catchment area for all women combined and also for white non-Hispanic, black, Hispanic, and Asian and Pacific Islander women; the small number of cases among American Indians ($n = 97$) precluded meaningful analysis of their data. For 1980 and 1990, analyses are based on the single race option provided in the 1980 and 1990 census; for 2000, they are based on persons reporting only one race, who comprised 97.4% of the US population. Exclusion of the 2.6% of persons in the 2000 census who reported more than one race, of whom approximately 60% were below age 30, [38] is unlikely to have biased our results, since only 0.5% of the study's breast cancer cases ($n = 332$) for 1998–2002 were for persons under 30 years old.

Geocoding of cases

To ensure consistency in geocoding across registries, which had previously independently geocoded their data, we had all cases geocoded by a commercial geocoding company whose accuracy we previously had tested and found to be high (96%) [39]. Comparison of geocodes assigned by the company and the registries indicated high agreement (94.7%) for the census tract geocodes for those cases ($n = 73{,}805$) geocoded to the high level of precision (exact street address or ZIP+4 code; the latter is an area typically the size of one city block). Our protocol for geocoding was as follows: (1) we used the geocoding company geocode (93.2% of total cases) if: (a) the geocoding company and the registry were both high-precision (assigned based on exact street address or ZIP+4 code) and the geocodes were identical (44.4% of total cases) or differed (2.5% of total cases), or (b) the geocoding company geocode was high-precision but the registry's was not (46.3% of total cases); (2) we used the registry geocode if it was high-precision and the geocoding company's was not (0.8% of total cases); and (3) if neither of the two geocodes

Table 1 Breast cancer incidence rate among women (per 100,000, age-standardized to the year 2000 million) by census tract poverty and high income and by race/ethnicity: San Francisco (SF) Bay Area and Los Angeles County (1978–1982, 1988–1992, 1998–2002), and Massachusetts (MA) (1988–1992, 1998–2002)

Area + group	Census tract poverty and high income	1978–1982						1988–1992						1998–2002					
		N	Person-years × 1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI
SF Bay Area																			
All women	<5% poverty and ≥10% high inc	487	389	116	(105,127)	1.23	(1.10,1.38)	2366	1364	153	(147,160)	1.40	(1.30,1.50)	4365	2377	156	(151,160)	1.53	(1.43,1.65)
	<5% poverty and < 10% high inc	2005	2007	107	(102,112)	1.14	(1.05,1.24)	3315	2370	138	(134,143)	1.26	(1.17,1.35)	2,406	1584	141	(136,147)	1.39	(1.29,1.51)
	5.0–9.9% poverty	3249	2917	107	(103,111)	1.14	(1.05,1.24)	4028	2825	137	(133,141)	1.25	(1.16,1.34)	4780	3409	128	(125,132)	1.27	(1.18,1.36)
	10.0–19.% poverty	1628	1684	100	(95,106)	1.07	(0.98,1.17)	1813	1617	120	(115,126)	1.10	(1.02,1.18)	2151	1981	116	(111,121)	1.15	(1.06,1.24)
	≥20% poverty	807	938	94	(87,101)	1.00		1088	1153	110	(103,117)	1.00		920	1064	101	(95,108)	1.00	
	Test for trend (*p*-value)					0.000						0.000						0.000	
White non-Hispanic women	<5% poverty ≥10% high inc	460	268	156	(142,172)	1.25	(1.07,1.45)	2109	1130	156	(150,163)	0.96	(0.86,1.08)	3667	1723	165	(160,171)	1.21	(1.06,1.38)
	<5% poverty, < 10% high inc	1793	1417	132	(126,138)	1.05	(0.93,1.19)	2736	1656	147	(141,152)	0.90	(0.81,1.01)	1778	883	158	(151,166)	1.16	(1.01,1.33)
	5.0–9.9% poverty	2801	2039	119	(115,124)	0.95	(0.84,1.08)	3161	1767	152	(146,157)	0.94	(0.84,1.04)	3310	1663	155	(150,160)	1.13	(0.99,1.30)
	10.0–19.% poverty	1185	844	128	(120,136)	1.02	(0.89,1.16)	1095	689	147	(138,157)	0.91	(0.80,1.02)	1140	672	144	(136,153)	1.05	(0.91,1.22)
	≥20% poverty	375	258	125	(111,141)	1.00		407	236	162	(146,180)	1.00		234	166	137	(119,156)	1.00	
	Test for trend (*p*-value)					0.001						0.368						0.000	
Black women	<5% poverty ≥10% high inc	5	7	72	(23,168)	0.82	(0.34,1.99)	37	26	139	(97,194)	1.19	(0.84,1.68)	54	47	105	(78,137)	0.90	(0.67,1.20)
	<5% poverty, < 10% high inc	33	55	99	(63,147)	1.13	(0.74,1.71)	87	102	109	(86,137)	0.93	(0.73,1.20)	77	73	120	(93,153)	1.03	(0.80,1.33)
	5.0–9.9% poverty	100	151	105	(83,130)	1.20	(0.94,1.53)	178	170	140	(120,163)	1.20	(1.00,1.43)	255	231	110	(97,125)	0.95	(0.81,1.11)
	10.0–19.% poverty	240	350	94	(82,107)	1.07	(0.90,1.28)	335	332	116	(104,129)	0.99	(0.86,1.14)	357	316	116	(105,129)	1.00	(0.87,1.15)
	≥20% poverty	311	436	88	(78,98)	1.00		475	469	117	(107,128)	1.00		412	373	117	(106,129)	1.00	
	Test for trend (*p*-value)					0.159						0.192						0.495	
Asian and Pacific Islander women	<5% poverty ≥10% high inc	8	21	38	(16,76)	0.73	(0.34,1.57)	127	146	111	(90,134)	2.03	(1.54,2.66)	407	388	106	(96,118)	1.46	(1.21,1.77)
	<5% poverty, < 10% high inc	66	165	53	(40,68)	1.01	(0.68,1.48)	267	358	93	(81,106)	1.70	(1.34,2.14)	304	316	101	(89,113)	1.38	(1.13,1.69)
	5.0–9.9% poverty	148	297	60	(50,70)	1.14	(0.82,1.59)	383	514	88	(79,98)	1.61	(1.30,2.01)	761	829	88	(82,95)	1.21	(1.02,1.44)
	10.0–19.% poverty	103	239	46	(37,56)	0.88	(0.62,1.24)	209	315	72	(63,83)	1.32	(1.04,1.68)	335	387	85	(76,94)	1.17	(0.96,1.42)
	≥20% poverty	50	100	52	(38,70)	1.00		111	216	55	(45,66)	1.00		157	214	73	(61,85)	1.00	
	Test for trend (*p*-value)					0.472						0.000						0.000	
Hispanic women	<5% poverty ≥10% high inc	9	13	85	(38,163)	1.17	(0.58,2.39)	78	61	150	(118,188)	2.07	(1.52,2.81)	171	146	138	(117,160)	1.87	(1.44,2.44)
	<5% poverty, < 10% high inc	86	173	77	(61,97)	1.07	(0.77,1.50)	216	256	120	(104,138)	1.65	(1.29,2.12)	225	247	132	(115,151)	1.80	(1.40,2.31)
	5.0–9.9% poverty	160	324	71	(60,83)	0.98	(0.73,1.32)	293	380	104	(93,117)	1.44	(1.14,1.82)	395	557	100	(90,111)	1.36	(1.08,1.72)
	10.0–19.% poverty	86	215	62	(50,77)	0.87	(0.62,1.20)	170	287	94	(80,110)	1.30	(1.00,1.68)	258	535	87	(76,98)	1.18	(0.92,1.51)
	≥20% poverty	66	133	72	(55,92)	1.00		96	241	72	(58,89)	1.00		96	277	73	(59,91)	1.00	
	Test for trend (*p*-value)					0.344						0.000						0.000	
Los Angeles County																			
All women	<5% poverty and ≥10% high inc	1019	841	118	(111,126)	1.51	(1.40,1.63)	3125	1732	157	(152,163)	1.72	(1.64,1.81)	3206	1581	164	(159,170)	1.79	(1.71,1.87)
	<5% poverty and < 10% high income	2634	2658	108	(103,112)	1.37	(1.30,1.45)	3500	2643	134	(130,139)	1.47	(1.40,1.54)	2118	1325	149	(143,156)	1.62	(1.54,1.71)
	5.0–9.9% poverty	5802	5438	105	(102,107)	1.34	(1.27,1.40)	7106	5376	131	(128,134)	1.44	(1.38,1.50)	7749	5134	138	(135,141)	1.50	(1.45,1.55)
	10.0–19.% poverty	4504	5553	90	(87,92)	1.14	(1.09,1.20)	6030	6300	114	(112,117)	1.25	(1.20,1.31)	7450	7012	115	(112,118)	1.25	(1.21,1.29)
	≥20% poverty	2440	4191	78	(75,81)	1.00		3765	6152	91	(88,94)	1.00		5800	9020	92	(90,94)	1.00	
	Test for trend (*p*-value)																		

Table 1. (Contd.)

Area + group	Census tract poverty and high income	1978–1982						1988–1992						1998–2002					
		N	Person-years × 1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI
Los Angeles County						0.000						0.000						0.000	
White	<5% poverty ≥10% high inc	955	727	124	(116,132)	0.85	(0.77,0.93)	2781	1351	165	(159,172)	1.16	(1.08,1.25)	2684	1122	177	(170,184)	1.19	(1.11,1.26)
non-	<5% poverty, < 10% high inc	2336	1971	119	(114,124)	0.82	(0.76,0.88)	2854	1758	144	(139,149)	1.01	(0.94,1.08)	1522	746	163	(155,171)	1.09	(1.02,1.17)
Hispanic	5.0–9.9% poverty	5140	3859	118	(115,122)	0.81	(0.76,0.87)	5605	3237	143	(139,147)	1.00	(0.94,1.07)	5478	2580	157	(153,162)	1.06	(1.00,1.12)
women	10.0–19.% poverty	3336	2478	118	(114,122)	0.81	(0.75,0.87)	3713	2140	140	(135,145)	0.98	(0.91,1.05)	4068	2100	147	(143,152)	0.99	(0.93,1.05)
	≥20% poverty	1090	692	146	(137,156)	1.00		1211	680	143	(134,152)	1.00		1697	923	149	(142,156)	1.00	
	Test for trend (*p*-value)					0.000						0.000						0.000	
Black	<5% poverty ≥10% high inc	10	16	91	(30,211)	1.13	(0.47,2.71)	44	40	121	(85,167)	1.15	(0.82,1.60)	92	55	145	(116,180)	1.21	(0.98,1.51)
women	<5% poverty, < 10% high inc	62	101	101	(73,135)	1.25	(0.92,1.69)	146	134	131	(109,155)	1.24	(1.03,1.49)	90	57	155	(124,193)	1.30	(1.04,1.62)
	5.0–9.9% poverty	136	255	88	(72,105)	1.09	(0.89,1.33)	356	347	136	(121,152)	1.29	(1.14,1.46)	346	279	128	(115,143)	1.07	(0.95,1.21)
	10.0–19.% poverty	463	777	85	(77,94)	1.06	(0.94,1.19)	723	825	114	(106,123)	1.08	(0.98,1.19)	798	709	120	(111,128)	1.00	(0.91,1.09)
	≥20% poverty	796	1291	81	(75,87)	1.00		1139	1268	105	(99,112)	1.00		1477	1384	120	(114,126)	1.00	
	Test for trend (*p*-value)					0.027						0.000						0.003	
Asian and	<5% poverty ≥10% high inc	12	39	33	(17,58)	1.21	(0.64,2.29)	144	204	81	(67,96)	1.36	(1.10,1.68)	241	222	103	(89,118)	1.34	(1.15,1.57)
Pacific	<5% poverty, < 10% high inc	49	178	38	(27,53)	1.43	(0.94,2.16)	205	332	72	(62,83)	1.21	(1.01,1.46)	245	225	103	(90,117)	1.34	(1.15,1.56)
Islander	5.0–9.9% poverty	68	297	31	(23,40)	1.14	(0.79,1.65)	395	618	77	(69,85)	1.30	(1.11,1.52)	867	800	104	(97,111)	1.36	(1.22,1.50)
women	10.0–19.% poverty	66	330	25	(19,32)	0.92	(0.64,1.33)	410	700	73	(66,81)	1.23	(1.05,1.43)	913	993	90	(84,96)	1.17	(1.06,1.30)
	≥20% poverty	58	243	27	(20,35)	1.00		287	583	59	(52,67)	1.00		624	806	77	(71,83)	1.00	
	Test for trend (*p*-value)					0.060						0.000						0.000	
Hispanic	<5% poverty ≥10% high inc	20	41	61	(35,97)	1.29	(0.79,2.10)	124	136	115	(94,138)	1.77	(1.45,2.15)	137	136	118	(98,140)	1.81	(1.51,2.18)
women	<5% poverty, < 10% high inc	104	314	54	(43,67)	1.15	(0.90,1.45)	260	421	94	(82,107)	1.45	(1.25,1.67)	235	257	121	(106,138)	1.86	(1.62,2.14)
	5.0–9.9% poverty	319	903	63	(55,71)	1.33	(1.14,1.56)	680	1180	90	(83,97)	1.39	(1.26,1.53)	952	1327	98	(92,105)	1.52	(1.40,1.64)
	10.0–19.% poverty	496	1855	50	(46,55)	1.07	(0.93,1.22)	1131	2666	78	(73,83)	1.20	(1.10,1.31)	1600	3024	84	(80,89)	1.30	(1.21,1.39)
	≥20% poverty	425	1923	47	(42,52)	1.00		1116	3710	65	(61,69)	1.00		1983	5769	65	(62,68)	1.00	
	Test for trend (*p*-value)					0.000						0.000						0.000	
Masachusetts																			
All women	<5% poverty and ≥10% high inc							1546	1361	103	(98,108)	0.95	(0.88,1.02)	3841	2237	150	(145,155)	1.35	(1.28,1.42)
	<5% poverty and < 10% high income							6790	5086	128	(125,131)	1.17	(1.11,1.24)	7269	4555	139	(136,142)	1.25	(1.19,1.31)
	5.0–9.9% poverty							6336	4770	118	(115,121)	1.09	(1.03,1.15)	7270	4782	130	(127,133)	1.17	(1.11,1.23)
	10.0–19.% poverty							2976	2441	114	(110,119)	1.05	(0.99,1.12)	3529	2800	120	(116,124)	1.08	(1.02,1.14)
	≥20% poverty							1680	1839	109	(103,114)	1.00		1987	2077	111	(106,116)	1.00	
	Test for trend (*p*-value)											0.020						0.000	
White non-	<5% poverty ≥10% high inc													3640	2071	149	(144,154)	1.14	(1.07,1.21)
Hispanic	<5% poverty, < 10% high inc													6982	4266	139	(136,142)	1.06	(1.00,1.12)
women	5.0–9.9% poverty													6861	4242	131	(128,134)	1.00	(0.94,1.06)
	10.0–19.% poverty													3052	1978	128	(123,132)	0.97	(0.91,1.04)
	≥20% poverty													1426	960	131	(124,138)	1.00	
	Test for trend (*p*-value)																	0.000	
Black	<5% poverty ≥10% high inc							6	13	63	(23,138)	0.80	(0.35,1.83)	17	26	70	(41,112)	0.68	(0.42,1.11)
women	<5% poverty, < 10% high inc							40	54	113	(80,155)	1.45	(1.02,2.05)	62	62	115	(88,148)	1.12	(0.85,1.49)
	5.0–9.9% poverty							78	128	85	(67,107)	1.09	(0.84,1.42)	111	136	102	(83,123)	0.99	(0.79,1.24)

Table 1 (Contd.)

Area + group	Census tract poverty and high income	1978–1982						1988–1992						1998–2002					
		N	Person-years × 1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI	N	Person-years ×1000	Rate	95% CI	IRR	95% CI
Masachusetts																			
	10.0–19.% poverty							95	183	75	(60,92)	0.96	(0.75,1.23)	233	317	92	(80,105)	0.90	(0.75,1.08)
	≥20% poverty							219	398	78	(68,89)	1.00		275	347	102	(91,115)	1.00	.
	Test for trend (*p*-value)											0.223						0.911	
Asian and	<5% poverty ≥10% high inc							5	36	25	(6,66)	0.73	(0.24,2.26)	52	83	72	(52,95)	1.69	(1.10,2.59)
Pacific	<5% poverty, < 10% high inc							21	71	49	(28,79)	1.44	(0.73,2.81)	53	98	72	(52,96)	1.69	(1.10,2.61)
Islander	5.0–9.9% poverty							17	88	33	(18,55)	0.97	(0.48,1.93)	60	156	55	(41,72)	1.29	(0.85,1.97)
women	10.0–19.% poverty							5	66	10	(3,26)	0.30	(0.10,0.89)	40	136	43	(30,60)	1.02	(0.65,1.62)
	≥20% poverty							19	101	34	(20,53)	1.00		40	139	42	(30,58)	1.00	
	Test for trend (*p*-value)											0.762						0.002	
Hispanic	<5% poverty ≥10% high inc													18	30	109	(63,176)	1.55	(0.93,2.57)
women	<5% poverty, < 10% high inc													35	69	88	(59,126)	1.24	(0.84,1.84)
	5.0–9.9% poverty													73	157	90	(69,115)	1.27	(0.95,1.69)
	10.0–19.% poverty													117	271	82	(66,101)	1.16	(0.90,1.50)
	≥20% poverty													209	558	71	(60,82)	1.00	.
	Test for trend (*p*-value)																	0.182	

were high precision, we categorized the case as ''un-geocodable'' (5.0% of total cases).

Calculation of census-derived area-based socioeconomic measures

We selected and constructed our census tract level area-based socioeconomic measures (ABSMs) based on theoretical considerations and methods described in detail in the publications of the *Public Health Disparities Geocoding Project* [21, 32]. ABSMs generated pertained to: poverty (% of population below poverty); income (median household income; % of population below 50% of median household income); occupation (% employed adults in working class occupations comprised predominantly of non-supervisory employees); education (% adults age 25 and older who were college graduate; % adults age 25 and older with less than a high school education); household crowding; Townsend deprivation index (the main area-based deprivation index used in the United Kingdom); a composite index combining data on income, education, and occupation used in prior analyses of cancer incidence rates in Southern California [16]; and a new composite variable we created combining data on poverty and high income, described below. We used census tract level data, rather than data from the block group or ZIP Code level, since results of our prior research indicated that ABSMs at the CT level yielded maximal geocoding and linkage to area-based socioeconomic data (compared to block group and ZIP Code data) and consistently detected expected socioeconomic gradients in health across a wide range of health outcomes [21, 32]. The proportion of the study catchment population living in CTs for which ABSM data were missing was small (ranging from a low of 0–0.3% for the 1990 and 2000 ABSMs to a high of 0.4–3.4% for the 1980 ABSMs) and did not vary by race/ethnicity.

Because results were overall robust to choice of ABSM, we present data for only the composite poverty+income ABSM, which combined data on the percent of persons below poverty and the percent of high income households (defined as ≥4 times the US median household income, and calculated from the categorical income distribution by interpolation, assuming a Pareto distribution within the income category [21, 32]). We modeled this ABSM as a 5-level categorical variable as follows: (a) among census tracts with < 5% poverty, we distinguished between those with < 10% and ≥10% high income households; and (b) among the remaining census tracts, we distinguished between those with 5.0–9.9%, 10.0–19.9%, and ≥20% poverty (the federal definition of a ''poverty area'') [21, 32, 40].

Calculation of age-standardized rates and rate ratiosfor each cross-sectional time period

To calculate age-specific incidence rates by stratum of CT ABSM, we aggregated cases (numerators) and person-time at risk (denominators) within each specific age stratum over all census tracts within a CT ABSM category. We computed age-standardized rates by CT ABSM by the direct method, using the year 2000 U.S. standard million population [37]. To avoid lower bounds that were less than zero, we calculated 95% confidence limits for the standardized rates using a method based on the gamma distribution [37, 41]. We estimated the age-adjusted incidence rate ratios by computing the ratio of the age-standardized incidence rates as calculated above. Within each time period, for each ABSM, we calculated the Mantel test for trend by socioeconomic position over age strata [42]. To determine if there were significant differences, within each racial/ethnic group, in the age-standardized socioeconomic incidence rate ratios (comparing the most to least affluent group) by time, comparing the earliest to latest time periods available (1980 versus 2000 for the California registries, 1990 versus 2000 for Massachusetts), we tested the equality of the log of the rate ratio. All data manipulation and calculations were conducted using SAS Version 9.0 [43].

Results

Sociodemographic composition of the catchment areas

The racial/ethnic and socioeconomic composition of the population in the three study catchment areas are shown in Figures 1 and 2. Between 1980 and 2000, all three areas experienced both increasing racial/ethnic diversity along with a rise in the proportion of the population born outside of the US (Figure 1). In all three catchment areas, and in all three time periods (1980, 1990, 2000) the black and Hispanic populations were most likely to live in poorer census tracts, while the white non-Hispanic population, followed by the Asian population, were most likely to live in the highest income census tracts, with this trend accelerating over time.

Breast cancer incidence rates stratified by socioeconomic position, by time, race/ethnicity, and gender

Table 1 presents data for each catchment area, for each time period, on the age-standardized breast cancer incidence rate by census tract socioeconomic level, overall and by race/ethnicity. Results for all women combined would appear to invalidate the study hypothesis: the socioeconomic gradient increased over time in all three catchment areas. For example, considering all women in Los Angeles county for the time periods 1978–1982 and 1998–2002, the point estimate of the incidence rate ratio comparing women living in census tracts with the most compared to least resources increased by 20% for the poverty + income ABSM (i.e., from 1.5 to 1.8; p-value = 0.001).

This overall pattern, however, differed sharply by race/ethnicity, whereby there was relatively little change over time in the magnitude of the socioeconomic gradient among white non-Hispanic and black women, but generally a marked increase among the Asian and Pacific Islander and Hispanic women. Thus, among white non-Hispanic women, the point estimate of the socioeconomic gradient for women in 1998–2002 was only about 1.2 in each of the three catchment areas, the same as observed in the San Francisco Bay Area for 1978–1982 (p-value for San Francisco compariso$n$ = 0.623). Among black women, the point estimate of the socioeconomic gradient likewise showed little change over time and in 1998–2002 ranged between 0.7 (Massachusetts) and 1.2 (Los Angeles County) (p-values for comparisons of 1978–1982 versus 1998–2002 incidence rate ratios all in excess of 0.4).

Conversely, among the Asian and Pacific Islander American women, the magnitude of socioeconomic gradient comparing the time periods 1978–1982 to 1998–2002 increased sharply in the San Francisco Bay Area (up by 200%, from 0.8 to 1.5; p-value = 0.042), rose incrementally (albeit non-significantly) in the Los Angeles area (from 1.2 to 1.35; p-value = 0.380), and tended to increase in Massachusetts (comparing 1988–1992 versus 1998–2002, from 1.0 to 1.4; p-value = 0.088). Among Hispanic women, the magnitude of the point estimate of the socioeconomic gradient tended to increase by 40–60% over time (p-values $\approx$ 0.10) and in 1998–2002 ranged from 1.6 (Massachusetts) to 1.9 (San Francisco Bay Area).

Discussion

Two key and novel findings emerge from our analyses of the socioeconomic gradient in breast cancer incidence over the last quarter of the 20th century, as manifested in three regions of the US: Northern and Southern California and Massachusetts. The first is that had we considered socioeconomic position without regard to race/ethnicity, we would have concluded that the magnitude of the socioeconomic gradient is increasing over time, with risk increasingly higher among women living in census tracts with more versus less socioeconomic resources. The second is that analyses stratified by race/ethnicity instead showed little change or even modest decline in the relatively small socioeconomic gradient evident among white non-Hispanic women and black women (incidence rate ratio in 1998–2002 $\approx$ 1.2), but provided evidence of a generally marked increase among the Asian and Pacific Islander and Hispanic women, for whom the socioeconomic gradient in 1998–2002 ranged from 1.3 to 1.9. Together, these results indicate considerable heterogeneity in the socioeconomic gradient in US breast cancer incidence and underscore the necessity of considering both socioeconomic position and race/ethnicity when monitoring and investigating the social patterning of disease within the US; neither alone is sufficient.

Before offering a detailed interpretation of our findings or comparing them to results of prior investigations, we first consider study limitations. It is unlikely that results were affected by systematic errors involving the assignment of geocodes, given the steps we took to ensure accuracy and precision of geocoding. Nor it is likely that results are biased by our reliance on ABSMs, rather than individual-level socioeconomic data, given substantial research indicating that aptly chosen ABSMs at the census tract level can be validly used to monitor socioeconomic gradients in health [21, 32]. Moreover, a very high proportion (95%) of the cases were geocoded and there was very little or no missing ABSM data. We likewise analyzed data for a wide array of ABSMs and results were robust to choice of ABSM.

We also note that while systematic errors may have occurred regarding the racial/ethnic data in both the numerator data (e.g., misclassification of race/ethnicity among cases) and the denominator data (e.g., differential population undercounts by race/ethnicity), [16, 32, 38, 44, 45] these problems would not have compromised comparisons of the socioeconomic gradient *within* the diverse racial/ethnic groups. Imputation of the white non-Hispanic counts by age and gender for the 1980 census, however, may have resulted in erroneous estimates of incidence rates among this group. We nevertheless decided to use the imputed rates because of the marked difference in the socioeconomic patterning of breast cancer incidence among white non-Hispanics and Hispanics in the other two time periods (1990 and 2000 census) for which we had access to counts of the white non-Hispanic population by age and gender. An additional limitation concerns the generalizability and precision of estimates: because our results are based on data from only two states, they cannot be generalized to all women in the US, and because of regional variation in rates, we opted not to combine data across the three catchment areas, thereby reducing the precision of the estimated incidence rates and rate ratios. Nor did the estimates take into account the potential spatial and temporal correlations of the data, which will be addressed in subsequent analyses. None of these limitations, however, would suggest the internal validity of our study is low.

The finding of at most a modest socioeconomic gradient among the US white non-Hispanic and black women, the two groups with the highest rates of breast cancer incidence in the US, is consistent with the initial study hypothesis, given higher socioeconomic gradients reported for these groups in earlier time periods [12–19]. The implication is that class differences in risk of breast cancer among these groups have narrowed considerably, likely for reasons described at the outset involving changes in reproductive history and anthropometry [1–12, 31]. Ideally, had it been possible to obtain and geocode US cancer registry data prior to the 1970s (i.e., prior to the establishment of a national system of cancer registries), we could have ascertained the periods of sharpest decline and onset of the subsequent plateau in the socioeconomic gradient.

By contrast, the generally – and in some registries, sharply – rising socioeconomic gradient among the US Asian and Pacific Islander and Hispanic women, groups whose absolute incidence rates are relatively low (compared to the US white non-Hispanic and black women), likely reflects a combination of demographic and health changes related to immigration. Of note, US breast cancer rates are currently among the highest documented by cancer registries worldwide [31]; by implication, rates of breast cancer incidence among immigrants to the US on average are lower compared to those of US-born women. Suggesting that differing societal contexts underlie these risk differentials, research indicates that the incidence of breast cancer among descendants of Asian and Hispanic immigrants soon catches up to that of the US-born women [46, 47]. Socially patterned changes in exposures thought to contribute to this change in risk include shifts in reproductive history (e.g., earlier age at menarche, reduced parity and later age at first pregnancy), diet (higher calorie), and body build (e.g., greater height and weight) [1, 2, 46, 47].

Also relevant to interpreting our study results is that, since the 1980s, the proportion of foreign-born among the US Asian and Pacific Islander and Hispanic populations has risen, and in 2000 was documented in the US census to be disproportionately concentrated in poorer census tracts [36] (data not shown; available upon request). One implication of the lower risk of breast cancer among immigrant women combined with the greater likelihood of immigrants to live in poorer CT would be to deflate the incidence rates in poverty areas relative to what they would be if restricted to only US-born Asian and Pacific Islanders and Hispanics, thereby increasing these groups' observed socioeconomic gradient in breast cancer incidence. Another implication would be that detection of the hypothesized decrease in the socioeconomic gradient would be possible only among US populations with low immigration, i.e., US white non-Hispanic and black women, which is in fact what we observed. Data limitations, however, render it impossible to test hypotheses about immigration effects directly using cancer registry data, due to limited data on the cases' birthplace. At issue is the decline of reporting of data on birthplace in medical records (the source of cancer registry data), due in part to rising anti-immigrant sentiment and legislation, especially in California [48]: in Northern California, the proportion of breast cancer cases with unknown birthplace in our study population rose, between 1973 and 2002, from 18% to 63%.

Suggesting our findings are not anomalous, other studies in both the US and Europe, as noted previously, when considered together, have indicated that the socioeconomic gradient in breast cancer may be declining over time, [12–31] in part because of changing patterns by birth cohort [28–31]. Despite suggestions these results may be due to improved screening among women with fewer socioeconomic resources, [15, 28] it is unlikely that only changes in screening could account for these trends, since cancer registries capture virtually all invasive cases, whether detected at an early stage or, far less commonly, at autopsy. Importantly, however, none of these prior investigations on socioeconomic gradients in breast cancer incidence investigated the impact of changing patterns of immigration on their results.

In conclusion, results of our study suggest that: (1) it is meaningless to test the hypothesis about changing socioeconomic gradients in US breast cancer incidence among the ''total population'' only, without also paying attention to race/ethnicity and immigration, and (2) within the US, the socioeconomic patterning of breast cancer incidence is not a unitary phenomenon but instead may vary across population groups, due to differences in social and demographic experiences. One manifestation of this heterogeneity would be the observed modest socioeconomic gradient stable or declining among the US white non-Hispanic and black populations in contrast to the markedly higher and rising gradient among the Asian and Pacific Islander and Hispanic populations. Related, our data underscore the danger of current efforts to prohibit state governments from collecting or using racial/ethnic data, as proposed by the recently defeated Proposition 54 ballot initiative in California [49]. Had we analyzed our data only by socioeconomic position, and not also by race/ethnicity, we would have reached the wrong conclusion. Finally, our results raise important questions about purported changes in US population health that do not take into account the likely impact of immigration on observed trends. Obtaining a ''truer picture'' [32, 50] of US breast cancer incidence, critical for guiding interventions and estimating the actual population burden of disease, will require considering the joint impact of socioeconomic position, race/ethnicity, and immigration status on population health.

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